

Horizon Logic (Structural Edition)

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Abstract

Horizon Logic formalises the behaviour of reasoning systems under bounded informational horizons. It provides the structural grammar governing horizon formation, horizon collapse, horizon opacity, and horizon drift. This edition presents the structural formulation: the basis-independent rules that define how horizons emerge, interact, and constrain inference.

1 Introduction

Horizon Logic describes how any reasoning system behaves when its inferential reach is limited by a structural horizon. A horizon is a boundary beyond which the system cannot reliably propagate inference, compression, or recurrence. This document presents the structural edition: the horizon grammar independent of any specific domain (mathematics, physics, AI reasoning engines, or simulation systems).

The structural edition defines:

- Horizon Formation
- Horizon Collapse
- Horizon Opacity
- Horizon Drift
- Horizon Interaction

2 1. Horizon Formation

A horizon forms when the system's structural information density exceeds its recurrence capacity. Let I denote information density and R recurrence capacity. A horizon forms when:

$$I > R.$$

The horizon radius H is defined structurally as:

$$H = f(I - R),$$

where f is a monotonic structural mapping.

2.1 1.1 Structural Conditions

A horizon forms when:

1. The system cannot maintain deterministic recurrence.
2. Compression gradients exceed structural stability.
3. Boundary information becomes non-propagating.

3 2. Horizon Collapse

Horizon collapse occurs when the system's structural recurrence operator fails to maintain horizon integrity. Collapse is defined by:

$$\Delta H < 0,$$

with collapse velocity:

$$v_c = -\frac{dH}{dt}.$$

Collapse is triggered by:

- Excessive recurrence load.
- Structural motif overload.
- Boundary instability.

4 3. Horizon Opacity

Opacity measures how much information is lost or distorted when crossing the horizon boundary. Structural opacity O is defined as:

$$O = 1 - \frac{I_{\text{out}}}{I_{\text{in}}},$$

where I_{in} is information inside the horizon and I_{out} is information that successfully propagates outward.

Opacity increases when:

- Compression gradients steepen.
- Recurrence becomes unstable.
- Boundary motifs become non-linear.

5 4. Horizon Drift

Drift describes the movement of the horizon boundary over time. Drift velocity v_d is defined structurally as:

$$v_d = \frac{dH}{dt}.$$

Drift occurs when:

- Information density changes.
- Recurrence capacity evolves.
- Boundary motifs reconfigure.

6 5. Horizon Interaction

Multiple horizons interact through structural interference. Interaction strength S is defined as:

$$S = g(H_1, H_2, d),$$

where d is the separation distance and g is a structural interaction mapping. Interactions include:

- Horizon merging.
- Horizon repulsion.
- Horizon resonance.

7 Conclusion

Horizon Logic provides the structural grammar governing bounded reasoning systems. It defines how horizons form, collapse, drift, and interact. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.